

Quantum Fault Tolerance Meets Toom's Contours

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May 26, 2026

1 Toric Encoded Circuits

Definition 1.1 (Toric Encoded Circuit). Let C be a quantum circuit. We say that C is a *Toric encoded circuit* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \dots, R_k$, and, at any time, each of the registers is encoded either by the *nice encoding of $\mathcal{C}$* or by the *nice encoding of $\phi \circ H(\mathcal{C})$* ¹. At time $t = 0$, the registers are initialized to be either $|\bar{0}\rangle$ or encoded $|\mathbf{S}\rangle$.
2. The layers of C at even times are sweep-cycles, and the layers at odd times are realizations of one of the logical $\mathbf{H}$, $\mathbf{S}$, $\mathbf{CX}$, $\mathbf{X}$, or $\mathbf{Z}$ gates.

Definition 1.2 (The Boundary of Deviation ∂D_t). Let C be a Toric encoded circuit, with registers $R_1, \dots, R_l$ and let $\mathcal{E}$ be a set of faults. Consider the subsection $C[\mathcal{E}]$. For a time t , let $A_t, B_t \subset \{R_1, \dots, R_l\}$ be the registers that encoded, at time t in $C[\mathcal{E}]$, by $\mathcal{C}$ and $\phi \circ \mathbf{H}$ respectively. Denote by $\mathcal{D}_t$ the deviation in $C[\mathcal{E}]$ at time t . We define the *X-type*, and the *Z-type boundaries* of $\mathcal{D}_t$, denoted by ∂D_t^X and ∂D_t^Z , as follow:

$$\begin{aligned}\partial \mathcal{D}_t^X &:= \bigcup_{R_i \in A_t} \partial |R_{i,t}\rangle \cup \bigcup_{R_i \in B_t} \partial (\phi \circ \mathbf{H}) |R_{i,t}\rangle \\ \partial \mathcal{D}_t^Z &:= \bigcup_{R_i \in B_t} \partial |R_{i,t}\rangle \cup \bigcup_{R_i \in A_t} \partial (\phi \circ \mathbf{H}) |R_{i,t}\rangle\end{aligned}$$

Definition 1.3 (Base graph G_o). Let C be a quantum circuit, at depth d , with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, \dots, \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C , denoted by $G_o = (V_o, E_o)$ as follows:

¹Notice that although the code space is invariant to the application of $\phi \circ \mathbf{H}$, the positive direction of the underlying Toric complex is flipped.

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \dots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$E_o := \{ \{(v, t), (u, t')\} : \text{if} \\ \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 1.4 (Tooms Contour Graph of $C(\mathcal{E})$). For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o . For each time t , denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial\mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the *Tooms contour graph* $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $V_{\mathcal{H}}$ are partitioned into two types, X -type vertices $V_{\mathcal{H},X}$ and Z -type vertices $V_{\mathcal{H},Z}$. Each of the X/Z -type corresponds to the vertices supporting the X/Z -boundary of $\mathcal{D}$. Namely:

$$V_{\mathcal{H},X} := \{v_t : v_t = (v, t) \in V_o \text{ and there exists a face } f \in \partial\mathcal{D}_t^X \text{ such } v \in f\} \\ V_{\mathcal{H},Z} := \{v_t : v_t = (v, t) \in V_o \text{ and there exists a face } f \in \partial\mathcal{D}_t^Z \text{ such } v \in f\}$$

Similarly, the Z -type vertices $V_{\mathcal{H},Z}$ are the vertices $(v, t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t , belongs to the $(\partial\mathcal{D}_t)|_Z$.

2. The edges $E_{\mathcal{H}}$ is partitioned into two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, where for any logical gate g which applied in time t in $C[\mathcal{E}]$ we add edge to $A_{\mathcal{H}}$ as follow:

- (a) If g is a logical **CX** then for any physical **CX** in its realization, between faces f_1 into f_2 at time t we add:

$$\{ \{(u, t), (v, t+1)\} : u \in f_1, v \in f_1 \cup f_2 \text{ and } v, u \in V_{\mathcal{H},X} \} \\ \{ \{(u, t), (v, t+1)\} : u \in f_2, v \in f_1 \cup f_2 \text{ and } v, u \in V_{\mathcal{H},Z} \}$$

- (b) If g is logical $\phi \circ \mathbf{H}$, then for any physical $\mathbf{H}$, in its realization, that acts on face f , we add:

$$\{ \{(u, t), (v, t+1)\} : u \in f, v \in \phi(f) \text{ and } v \in V_{\mathcal{H},Z}, u \in V_{\mathcal{H},X} \} \\ \{ \{(u, t), (v, t+1)\} : u \in f, v \in \phi(f) \text{ and } v \in V_{\mathcal{H},X}, u \in V_{\mathcal{H},Z} \}$$

- (c) If g is one of logical X, Z, I then for any face f , in the register on which acts g , we add:

$$\begin{aligned} & \{ \{(u, t), (v, t + 1)\} : u, v \in f \text{ and } u, v \in V_{\mathcal{H}, X} \} \\ & \{ \{(u, t), (v, t + 1)\} : u, v \in f \text{ and } u, v \in V_{\mathcal{H}, Z} \} \end{aligned}$$

- (d) If g is a sweep-cycle f , in the register on which acts g , we add:

$$\begin{aligned} & \{ \{(u, t), (v, t + 1)\} : u, v \in f \text{ and } u, v \in V_{\mathcal{H}, X} \} \\ & \{ \{(u, t), (v, t + 1)\} : u, v \in f \text{ and } u, v \in V_{\mathcal{H}, Z} \} \end{aligned}$$

In addition for any fault g of $\mathcal{E}$, that is applied at time t , and any face $f \in \text{Sup}(g)$, we add the edges:

$$\begin{aligned} A_{\mathcal{H}} &= \{ e = \{u, v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1 \} \\ F_{\mathcal{H}} &= \{ e = \{u, v\} \in E_o : \text{exists } w \in V_{\mathcal{H}} \text{ such } \{v, w\}, \{u, w\} \in A_{\mathcal{H}} \} \end{aligned}$$

By definition of the construction, $G_{\mathcal{H}}$ is a time-graph.

Definition 1.5 (Time Graph of $C(\mathcal{E})$). For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o . For each time t , denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial\mathcal{D}_t$ is the syndrome that the qubits in the deviation admit, denote by $\partial\mathcal{D}_t^X$ and by $\partial\mathcal{D}_t^Z$ the $X - Z$ decomposition of $\partial\mathcal{D}_t$. We define the history graph $\mathcal{H} = (V_{\mathcal{H}, X} \cup V_{\mathcal{H}, Z}, E_{\mathcal{H}})$, in recursive manner as follow:

1. For the empty circuit. The vertices in $V_{\mathcal{H}, X}$ are the vertices of $V_o|_1$ that support $\partial\mathcal{D}_1^X$. Similarly, the vertices of $V_{\mathcal{H}, Z}$ are the vertices of $V_o|_1$ that support $\partial\mathcal{D}_1^Z$.
2. Let $\mathcal{H}$ be the history graph of $C[\mathcal{E}]$ restricted to the first $d - 1$ layers. Let U be the d -layer, then define the history graph corresponding to the first d layers of $C[\mathcal{E}]$ to be obtained by adding vertices and edges to $\mathcal{H}$ as follows:
 - (a) If U is a noise gate, namely $U \subset \mathcal{E}$, then denote by $\mathcal{E}_t^X$ and $\mathcal{E}_t^Z$ the X, Z decomposition of $\mathcal{E}_t$. Then, for any vertex in $\mathcal{G}$
 - (b) Else, denote by $U_1, \dots, U_l$ the gates in U . If U_1 is a $\mathbf{CX}^{\otimes n}$ gate acting on registers R_1 into R_2 , then for each vertex $v \in V_{\mathcal{H}, X}|_{R_1, t=d-1}$ add the counter parts of v in R_1, R_2 at time $t = d + 1$ into $V_{\mathcal{H}, X}$ and connect them by arrows to v . Similarly, for any vertex $v \in V_{\mathcal{H}, Z}|_{R_2, t=d-1}$ add the counter parts of v in R_2, R_1 at time $t = d + 1$ into $V_{\mathcal{H}, X}$ and connect them by arrows to v .

[COMMENT] What we have to do is take the first definition, but split the vertex into X -type and Z -type. Then we connect the boundary of two registers in time if the gate applied is I or $\mathbf{CX}$; otherwise, in case we apply Hadamard, we cross the type. If we apply a Sweep cycle, then again we connect only vertices of the same type.